

1. System Overview

The Meal Tracker application is a basic fullstack web application that allows users to store and view meals. The goal of the app is to make it easy to track food in a structured way.

The application is built using React for the frontend and Node.js with Express for the backend. The backend handles API requests and communicates with the database.

MongoDB Atlas is used to store the data. The app is mainly intended as a learning project to understand how frontend, backend and database work together.

2. Database Design

The database consists of three collections: User, Food and MealLog.

The MealLog collection is the most important one because it connects the other two. It stores information about a meal and includes references to both User and Food using ObjectId.

This means that:

- one user can have multiple meals
- one food item can be used in multiple meals

This structure makes it easier to manage data and creates relationships between collections.

3. API Endpoints

Endpoint 1

- Route: `/api/meals`
- Method: GET
- Description: Returns all meals

Example response:

```
[
  {
    "_id": "123",
    "quantity": 2,
    "food": {
      "name": "Chicken"
    }
  }
]
```

Endpoint 2

- Route: /api/meals
- Method: POST
- Description: Creates a new meal

Example request:

```
{  
  "quantity": 2  
}
```

Example response:

```
{  
  "_id": "124",  
  "quantity": 2  
}
```

4. Reflection

One problem I had was connecting MongoDB Atlas to my backend. At first it did not work because my IP address was not allowed.

I fixed this by adding my IP address in the Network Access settings in MongoDB Atlas. After that, the connection worked.

This helped me understand how external databases work and how important configuration is.

5. Feature Iteration

One thing that changed during development was the MealLog model.

At the beginning, it only stored simple data like quantity. Later, I updated it to include references to User and Food.

This can be seen in the Git commits:

- 38c7256
- c5402a3

This change made the app more realistic and better structured.